

The Third DIMENSION

When was the last time you saw an innovation in the field of displays? It's the same, boring CRTs and LCDs. Right? Actually, researchers are working on technologies that will banish 2D from our lives forever!

Ram Mohan Rao

The world is 3D. But everything that's a reproduction of it—newspapers, TVs, computer displays—we watch in 2D. It's surprising the way we've gotten used to viewing and working with all this 2D. Have you ever, while playing a video game, craned your neck to the left to try and see what was *behind* something on the screen? Did you feel foolish about it later?

Think about 3D TV for a while: your favourite TV serial recorded and broadcast in true 3D. Imagine your family sitting all around the TV box, instead of in front of it, or perhaps even walking round it to get different perspectives. Sounds far fetched? Science fiction, right? Well, not exactly!

Yes, there's still quite a while to go before we'll be able to buy something like that off the shelf. Some say it'll be as soon as the year 2020, 'the year of perfect vision', that we'll get true real-time 3D displays you can walk all around. Others aver it'll be at least a fifty years from now. The truth is, no one knows for sure!

Our aim here is to inform you of some pretty amazing prototypes out there—devices that are the state of the art in holographic and stereoscopic techniques. And all developed and showcased today—not 50 years hence!

The Technologies

There are many 3D display technologies being worked upon. Three prominent categories are stereoscopic, holographic-in-air, and enclosed volumetric displays. Stereoscopic displays may require special glasses, IMAX-style, or may not, but they're basically just two different shots of a scene, each fed to one eye, to create the illusion of 3D.

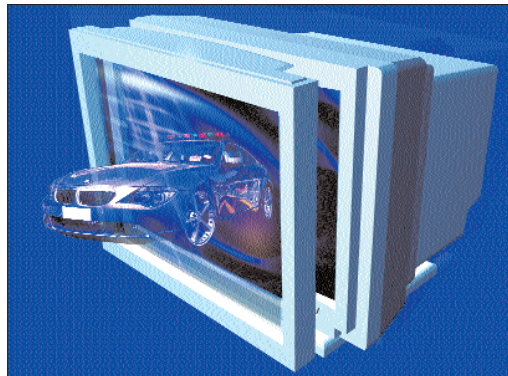
The term 'holography' means a lot of different things, but it's been popularised in movies such as Star Wars, with R2-D2 projecting the image of Princess Leia—so think about holography as 'the Star Wars' thing'!

Enclosed volumetric displays are probably the least known of the three—these are 3D renditions of scenes within a spinning volume. We take a look at examples of each of these.

Stereoscopic Displays

These are more advanced than you might imagine. Take a look at NuVision's Web site (www.nuvision3d.com): "Welcome. To the most advanced stereoscopic technology available. To absolute, true depth, flicker-free 3D realism. To a world as real as the one you live in. Welcome to the future."

One of their products—the SX line of display kits—uses a "high-speed liquid crystal modulator" that you use as an overlay for your existing monitor, in conjunction with special glasses, to give you 3D realism. Now if only this system were cheap enough for the majority to afford!



Monitor overlays, in conjunction with special glasses, can afford a degree of 3D realism

